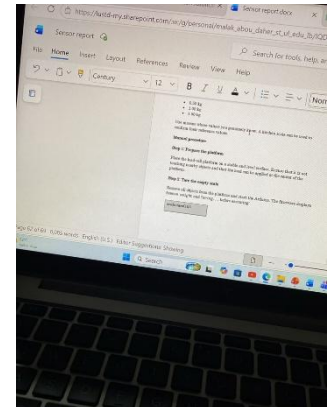


Imane

Technical Report, Data Organization, Analysis, and Documentation

Imane was mainly responsible for preparing and organizing the technical report and the recorded project data. She developed the written explanation of the project, ensured that the report followed the required structure, and combined the technical contributions of the team into one consistent document.



Her report work included the introduction, problem statement, system architecture, sensor selection and justification, hardware design, design evolution, software architecture, LCD-screen descriptions, calibration procedures, experimental results, error analysis, engineering constraints, conclusion, and future improvements.

Imane organized the component and pin-connection tables and prepared explanations of each sensor's measurement function, interface, role, selection reason, and limitations. She also documented the removal of the unreliable rain sensor, the replacement of rain-based control with BH1750 light control, and the unsuccessful SD-card logging attempt. This ensured that the report reflected the actual development and troubleshooting process rather than presenting only the final successful design.

She was also responsible for organizing the data exported from the Arduino. The Serial output was saved in Weather_Data.csv and prepared for use in Excel. Imane arranged the CSV fields, checked column headings, selected suitable measurement samples, and prepared tables and graphs for the experimental-results and calibration sections.

Time	Temp	Hum	Humidity	Light	Air Quality	Pressure	Altitude	Weight	Light	Total	Weather	Pressure	Light	Time	Status
1.12	25.7	77.8	10	100	100.1	100.1	0	0	0	0	0	0	0	0	0
3.13	25.8	76.6	77.5	105	100.1	100.1	0.047	0	0	0	0	0	0	0	0
5.14	25.9	75.6	10.8	102	100	100	0.047	0.04	0	0	0	0	0	0	0
7.15	26	75.5	130	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
9.16	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
11.17	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
13.18	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
15.19	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
17.20	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
19.21	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
21.22	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
23.23	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
25.24	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
27.25	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
29.26	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
31.27	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
33.28	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
35.29	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
37.30	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
39.31	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
41.32	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
43.33	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
45.34	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
47.35	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
49.36	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
51.37	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
53.38	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
55.39	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
57.40	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
59.41	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
61.42	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
63.43	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
65.44	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
67.45	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
69.46	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
71.47	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
73.48	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
75.49	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
77.50	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
79.51	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
81.52	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
83.53	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
85.54	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
87.55	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
89.56	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
91.57	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
93.58	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
95.59	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
97.60	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
99.61	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
101.62	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
103.63	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
105.64	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
107.65	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
109.66	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
111.67	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
113.68	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
115.69	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
117.70	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
119.71	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
121.72	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
123.73	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
125.74	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
127.75	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
129.76	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
131.77	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
133.78	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
135.79	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
137.80	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
139.81	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
141.82	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
143.83	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
145.84	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
147.85	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
149.86	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
151.87	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
153.88	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
155.89	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
157.90	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
159.91	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
161.92	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
163.93	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
165.94	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
167.95	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
169.96	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
171.97	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
173.98	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
175.99	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
178.00	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
180.01	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
182.02	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
184.03	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
186.04	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
188.05	26.1	75.3	10.8	104	100.1	100.1	0.048	0.04	0	0	0	0	0	0	0
190.06	26.1	75.3	10.8												

Her responsibilities also included inserting and captioning photographs, schematics, flowcharts, code listings, LCD screenshots, calibration figures, and prototype images. She reviewed the report for consistent terminology, numbering, formatting, and correspondence between the written explanations and the final firmware.



The project proposal requires the final report to discuss calibration, experimental results, uncertainty, error analysis, and engineering constraints; Imane's documentation work ensured that these requirements were addressed systematically.

Personal reflection:

Through preparing the report and organizing the data, I learned how to convert practical engineering work into clear technical documentation. I improved my ability to explain sensor operation, firmware logic, calibration procedures, design changes, and experimental results. I also learned the importance of using real evidence, consistent units, clear figures, and accurate tables so that another person can understand and reproduce the project.

